

# VASU BANSAL

Double Major - Mechanical Engineering · Aerospace Engineering · Indian Institute of Technology Kanpur

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## EXPERIENCE

- **MTS-3, Nutanix** May '25 – Present  
*NDB - R&D: IAM (Java, Go, Kubernetes, Helm, x.509 certificates)*
  - **Project Adrasteia Authentication** - NDB's DB On Container's initiative
    - ◇ **IAM Migration: Leading migration** of legacy IAM framework to an OIDC-compliant Kubernetes stack.
    - ◇ Did POCs, **collaboration** with multiple external teams, feat management.
    - ◇ Introducing a **two-cookie** approach to support migration.
    - ◇ **Mentored** an intern and bought him into speed with K8s and Helm.
  - Fixing security vulnerabilities and RBAC bugs.
- **SDE L2, GoTo Financial (Gojek)** June '22 – May '25  
*FinTech - Data Products & Infra Engineering - OneKYC (Backend - Java, Golang)*
  - **GCS Cloud Optimization** - Resulted in 25% reduction in GCS costs.
    - ◇ **Kubernetes Resource Efficiency**: Analyzed and adjusted CPU and memory requests and limits for various services running in K8s.
    - ◇ **JVM fine-tuning**: Fine-tuned `JAVA_OPTS` to optimize old-generation garbage-collection, resulting in better memory utilization and reduced latency.
  - **Partner Initiatives** - Entities which use our KYC process
    - ◇ Onboarded Gojek Driver, Vehicle Financing, Offline Merchants as partners
    - ◇ Cross-data pollination between Cashloan and Cicil
  - **PII Encryption** - For multiple services and respective databases
    - ◇ Planned **DB migrations** and dealt with edge-cases due to legacy data
    - ◇ Developed asynchronous workers for encryption and consistency jobs
  - **Analytics Service** - Uploading log data resulting in easier debugging
    - ◇ Generates on-demand presigned URLs for uploading data to GCS
    - ◇ Integrated **Kafka** events for BI consumption and data retrieval
  - Dedupe automation → **12%↑ in automation rate and 50%↓ in p90 SLA**
- **Analyst, Deloitte USI** June '21 – June '22  
*Project CogSpend - Supply Chain & Network Operations*
  - Automated file downloading for CogSpend database using **Python script**
  - Feature development in Springboot as per client requirements with unit-tests

## PROJECTS

- **Homelab**   April '24 - Present  
*Self motivated project - Work In Progress*
  - **Self-hosting** various services on Synology DS-923+ and RaspberryPi
    - ◇ **Adguard** for DNS blocking, **Vaultwarden** as password manger.
    - ◇ **Immich** as private photo backup and cloud service.
    - ◇ **Prometheus**, **Grafana**, **NodeExporter**, **cAdvisor** for monitoring.
    - ◇ Simple RAG system using **LangChain** + **Qdrant** to serve natural language answers over personal knowledge base<sup>a</sup>
- **Echo-Operator**   Oct '25 - Nov '25  
*Self motivated project*
  - **Kubernetes Custom Controller** to learn about CRDs
    - ◇ K8s operator using **controller-runtime** to manage EchoServer CRs.
    - ◇ Manages Deployment and Service objects based on custom-resource specs
    - ◇ Implemented CRD with validation, reconciliation loops, and status updates
- **Inventory Management**   Jan '23 - Feb '23  
*Self motivated project - MySQL, MongoDB, Containerization*
  - Implemented various modern micro-service paradigms listed below:
    - ◇ Inter-service communication, fault-tolerance using **Resilience4J**
    - ◇ Service discovery using **Eureka** and API Gateway using **Spring Cloud**
    - ◇ AuthN/AuthZ using **Keycloak**, distributed tracing using **Sleuth/Zipkin**
    - ◇ **Kafka** message queues, **ELK stack** for logs, **Prometheus** for monitoring

<sup>a</sup> work in progress

## EDUCATION

- **IIT Kanpur** July '16 - July '21  
*B.Tech. — CGPA - 7.8/10*
- **St. Kabir Convent School** March '16  
*AISSCE — CBSE Class XII — 88.4%*
- **St. Xavier's School** March '14  
*AISSE — CBSE Class X — 10 CGPA*

## ACADEMIC ACHIEVEMENTS

- Awarded **Academic Excellence Award** in 4th year at IIT Kanpur.
- JEE Advanced 2016: **AIR 1033**
- Qualified **RMO** and attended state-level workshop for INMO<sup>a</sup> 2015.
- NSTSE<sup>b</sup> 2014: **State Rank 56**

## TECHNICAL SKILLS

- **Programming Languages:**  
Java, Golang, C++, Python, Bash
- **Tech Stack:**  
Springboot, RoR<sup>c</sup>, Node.JS, React.JS
- **Software and Utilities:**  
PostgreSQL, Kafka, Redis, Docker, Kubernetes, Helm, Istio, Elasticsearch, Envoy, Git, gRPC, Postman

## LINKS

- Active on Leetcode, Codechef (3★), Codeforces (Rating - 1165)

## RELEVANT COURSES

- Data Structures & Algorithms
- Machine Learning
- Autonomous Navigation
- Robot Manipulators
- Robot Motion Planning
- Optimization Theory
- Advanced Statistical Methods
- Numerical Methods
- Finite Element Methods

### Online Courses:

- Completed **HTTP Server, DNS server and BitTorrent client** challenges in Go on Codecrafters
- Deep Learning Specialization (Coursera)

<sup>a</sup> Indian National Mathematics Olympiad

<sup>b</sup> National Science Talent Search Examination

<sup>c</sup> Ruby on Rails